

Section 3 — SQL File

Raw SQL reference (schema + queries). Full explanation and results are in answers.docx.

Schema

```
CREATE TABLE customers (  
    customer_id INTEGER PRIMARY KEY,  
    customer_name VARCHAR(100) NOT NULL,  
    customer_segment VARCHAR(20) NOT NULL CHECK (customer_segment IN ('Individual',  
'Group', 'Corporate')),  
    customer_signup_date DATE NOT NULL,  
    customer_home_city VARCHAR(50),  
    customer_loyalty_tier VARCHAR(20) NOT NULL DEFAULT 'None'  
    CHECK (customer_loyalty_tier IN ('None', 'Silver', 'Gold', 'Platinum'))  
);  
  
CREATE TABLE properties (  
    property_id INTEGER PRIMARY KEY,  
    property_name VARCHAR(100) NOT NULL,  
    property_city VARCHAR(50) NOT NULL,  
    property_star_rating INTEGER CHECK (property_star_rating BETWEEN 1 AND 5),  
    property_type VARCHAR(30) NOT NULL,  
    property_total_rooms INTEGER CHECK (property_total_rooms > 0),  
    UNIQUE (property_name, property_city, property_id)  
);  
  
CREATE TABLE bookings (  
    booking_id INTEGER PRIMARY KEY,  
    customer_id INTEGER NOT NULL REFERENCES customers(customer_id),  
    property_id INTEGER NOT NULL REFERENCES properties(property_id),  
    booking_date DATE NOT NULL,  
    checkin_date DATE NOT NULL,  
    checkout_date DATE NOT NULL,  
    room_type VARCHAR(30),  
    num_rooms INTEGER NOT NULL CHECK (num_rooms > 0),  
    nights INTEGER NOT NULL CHECK (nights > 0),  
    booking_channel VARCHAR(20) NOT NULL  
    CHECK (booking_channel IN ('Direct Website', 'OTA', 'Travel Agent', 'Corporate  
Portal')),  
    adr DECIMAL(10,2) NOT NULL CHECK (adr >= 0),  
    discount_amount DECIMAL(10,2) DEFAULT 0 CHECK (discount_amount >= 0),  
    coupon_code VARCHAR(20),  
    total_amount DECIMAL(12,2) NOT NULL CHECK (total_amount >= 0),  
    payment_method VARCHAR(30),  
    booking_status VARCHAR(20) NOT NULL  
    CHECK (booking_status IN ('Completed', 'Cancelled', 'No-Show')),  
    CHECK (checkout_date > checkin_date)  
);  
  
CREATE TABLE reviews (  
    review_id INTEGER PRIMARY KEY,  
    booking_id INTEGER NOT NULL UNIQUE REFERENCES bookings(booking_id),  
    review_rating DECIMAL(3,1) NOT NULL,  
    review_date DATE NOT NULL  
);
```

Indexes

```
CREATE INDEX idx_bookings_property_status ON bookings(property_id, booking_status);  
CREATE INDEX idx_bookings_customer_checkin ON bookings(customer_id, checkin_date);
```

Queries

A-Q1 — Highest Realized Revenue per City

```
WITH property_revenue AS (  
    SELECT  
        property_id,  
        property_name,  
        property_city,  
        SUM(total_amount) AS revenue  
    FROM bookings_flat  
    WHERE booking_status = 'Completed'  
    GROUP BY property_id, property_name, property_city  
)  
,  
ranked AS (  
    SELECT  
        property_id,  
        property_name,  
        property_city,  
        revenue,  
        RANK() OVER (PARTITION BY property_city ORDER BY revenue DESC) AS rnk  
    FROM property_revenue  
)  
SELECT property_city, property_id, property_name, revenue  
FROM ranked  
WHERE rnk = 1  
ORDER BY property_city;
```

Dialect: SQLite, executed against bookings_flat (flat CSV table). Result: 10 rows (one per city). Full result table is in answers.docx.

A-Q2 — Customers with Average Booking Gap Under 30 Days

```
WITH completed_ordered AS (  
    SELECT  
        customer_id,  
        checkin_date,  
        LAG(checkin_date) OVER (PARTITION BY customer_id ORDER BY checkin_date) AS  
prev_checkin  
    FROM bookings_flat  
    WHERE booking_status = 'Completed'  
)  
,  
gaps AS (  
    SELECT  
        customer_id,  
        julianday(checkin_date) - julianday(prev_checkin) AS gap_days  
    FROM completed_ordered  
    WHERE prev_checkin IS NOT NULL  
)  
,  
customer_avg_gap AS (  
    SELECT  
        customer_id,  
        AVG(gap_days) AS avg_gap  
    FROM gaps  
    GROUP BY customer_id  
)  
SELECT COUNT(*) AS customers_under_30_day_avg_gap  
FROM customer_avg_gap  
WHERE avg_gap < 30;
```

Dialect: SQLite — julianday() used for date subtraction (no native DATE arithmetic operator). Result: 445 customers.